

Zeravuln – Web Vulnerability Scanner

Introduction

Zeravuln is a Python-based web vulnerability scanner designed to identify common and critical security flaws in web applications. Built with extensibility and simplicity in mind, the tool leverages crawling, payload injection, and signature-based detection to find issues like SQLi, XSS, LFI, open redirects, and command injection vulnerabilities.

Abstract

The Zeravuln project automates the task of identifying web security vulnerabilities in target applications. It accepts a URL, scans pages and forms, injects test payloads, and analyzes responses to detect security flaws. Reports are generated in JSON and HTML formats for further analysis.

Tools Used

- Python 3.10
- Requests
- BeautifulSoup (bs4)
- HTML & JSON for reporting
- Git & GitHub for version control

Steps Involved in Building the Project

1. Designed the project architecture and named it Zeravuln.
2. Developed a recursive crawler to explore links and forms.
3. Created and loaded payload sets for each vulnerability type.
4. Implemented vulnerability detection functions for SQLi, XSS, LFI, CMDi, and open redirect.
5. Handled form submission logic with payload injection.
6. Built a reporting module for HTML and JSON output.
7. Added verbosity levels and argparse CLI arguments.

8. Tested the tool on vulnerable websites and adjusted detection accuracy.
9. Published the tool on GitHub.

Conclusion

Zeravuln provides an effective and extensible approach to automated web vulnerability scanning. It can be expanded to detect more complex vulnerabilities and integrated into broader security assessment pipelines. The tool enhances awareness of web application security and supports secure development practices.